YEAR 10 PHYSICS, AQA GCSE, PART 3 OF 3

PARTICLE MODEL OF MATTER

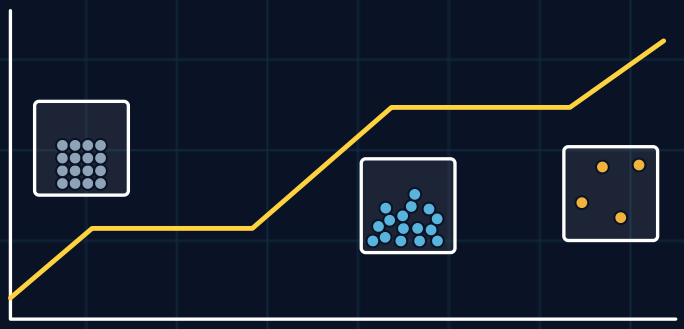
PART 3 OF 3

LESSONS 6–7

IN THIS BOOKLET

- Lesson 6: Specific latent heat
- Lesson 7: Specific heat capacity

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

HOW THIS BOOKLET WORKS

This is the booklet for **Lessons 6–7** of the Particle Model of Matter unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep the booklet: it is your classwork book *and* your revision guide.

DO NOW

Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT

Explanations and diagrams. Fill in the slide examples with your teacher, then try “Your turn” alone.

TRY IT

Practice that builds up, then a challenge. Every second lesson ends with the slides' Quick check.

PRACTICAL

Record your measurements and work out results from them.

EXAM CORNER

AQA-style questions with a tip on how the marks are given.

REVISE IT

Summary, cover-and-answer questions and an “I can” check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The particles in a solid are close together **(close together / far apart)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of mass is **the newton** | **the kilogram** | **the joule** .

THE ROUTE THROUGH THE UNIT (THIS BOOKLET HIGHLIGHTED)

01 Solids, liquids and gases

02 Gas particles and pressure

03 Density

04 Measuring density (required practical)

05 Internal energy and heating curves

06 Specific latent heat
Slides 29–34

07 Specific heat capacity
Slides 35–39

08 Measuring specific heat capacity (required practical)

The **equation toolkit** near the back has every equation you have met so far, with formula triangles to help you rearrange. Teaching deck: **panphy.app/gcsephy/decks/particle-model**

Lesson

06 SPECIFIC LATENT HEAT

AQA Physics 4.3.2.3; Combined Science 6.3.2.3

BIG QUESTION **A scald from steam at 100 °C is far worse than a scald from water at 100 °C. Why?**

BY THE END YOU CAN

- state what specific latent heat means
- calculate with $E = m\,L$
- tell fusion from vaporisation
- describe an experiment to measure the specific latent heat of fusion of water

latent heat **specific latent heat** **fusion** **vaporisation** **joule per kilogram (J/kg)**

YOU WILL NEED

Per group, or as a demonstration: two funnels of crushed ice over beakers, 12 V immersion heater, joulemeter, power supply, balance, stopwatch.

Teaching deck, slides 29–34
panphy.app/gcsephy/decks/particle-model/#29

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Define internal energy.

The total kinetic and potential energy of all the particles in a system.

2
What happens to the temperature of a substance while it changes state?

It stays constant.

3
Convert 250 g to kilograms.

0.250 kg

EXPLAIN

THE HIDDEN ENERGY

The energy needed for a substance to change state is called **latent heat**. “Latent” means hidden: the energy goes in, the internal energy increases, but a thermometer shows no change.

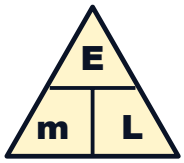
! The **specific latent heat** of a substance is the amount of energy required to change the state of **one kilogram** of the substance with **no change in temperature**.

$$E = m\,L$$

energy for a change of state = mass × specific latent heat

E in J, m in kg, L in J/kg

Rearranged: $L = \frac{E}{m}$ and $m = \frac{E}{L}$



TWO KINDS

	CHANGE OF STATE	VALUE FOR WATER
Specific latent heat of fusion	solid to liquid (melting), or liquid to solid (freezing)	334 000 J/kg
Specific latent heat of vaporisation	liquid to vapour (boiling), or vapour to liquid (condensing)	2 260 000 J/kg

Boiling needs much more energy than melting, because the particles must be pulled completely apart. When the change goes the other way, the same amount of energy is **given out**. Steam that condenses on your skin transfers 2 260 000 J for every kilogram, *before* it starts to cool as hot water. That is why steam scalds so badly.

WITH YOUR TEACHER — SLIDE 31

Calculate the energy needed to melt 0.50 kg of ice at 0 °C. Specific latent heat of fusion of water = 334 000 J/kg

EQUATION $E = m\,L$

SUBSTITUTE $E = 0.50 \times 334\,000$

ANSWER $E = 167\,000$ J

YOUR TURN

Calculate the energy needed to boil away 0.20 kg of water at 100 °C. Specific latent heat of vaporisation of water = 2 260 000 J/kg

EQUATION $E = m\,L$

SUBSTITUTE $E = 0.20 \times 2\,260\,000$

ANSWER $E = 452\,000$ J

WITH YOUR TEACHER SLIDE 31

A heater transfers 90 000 J to a block of wax at its melting point. The specific latent heat of fusion of the wax is 150 000 J/kg. Calculate the mass of wax that melts.

EQUATION	$E = m\,L$
REARRANGE	$m = \dfrac{E}{L}$
SUBSTITUTE	$m = \dfrac{90\,000}{150\,000}$
ANSWER	**m = 0.60 kg**

YOUR TURN

It takes 50 400 J to melt 0.40 kg of a metal at its melting point. Calculate the specific latent heat of fusion of the metal.

EQUATION	$E = m\,L$
REARRANGE	$L = \dfrac{E}{m}$
SUBSTITUTE	$L = \dfrac{50\,400}{0.40}$
ANSWER	$L = 126\,000$ J/kg

Get the marks: There is no temperature in this equation. If a question gives a temperature change, you need a different equation (Lesson 7).

PRACTISE

TRY IT

1 Draw a ring around the correct answer in each box.

The unit of specific latent heat is **J** | (**J/kg**) | **J/kg °C**

Ice melting uses the specific latent heat of (**fusion**) | **vaporisation**

Steam condensing uses the specific latent heat of **fusion** | (**vaporisation**)

2 SUPPORTED Calculate the energy needed to melt 1.5 kg of ice at 0 °C. Specific latent heat of fusion of water = 334 000 J/kg

EQUATION	$E = m\,L$	
SUBSTITUTE	$E = 1.5 \times 334\,000$	
ANSWER	**501 000 J**	unit:

3 SUPPORTED A kettle boils away 150 g of water at 100 °C. Calculate the energy transferred to the water. Specific latent heat of vaporisation of water = 2 260 000 J/kg

Convert the mass first!

CONVERT	**m = 150 ÷ 1000 = 0.150 kg**
EQUATION	$E = m\,L$
SUBSTITUTE	$E = 0.150 \times 2\,260\,000$
ANSWER	**339 000 J** unit:

4 Complete the table. Show your working underneath.

Change of state	Mass / kg	Specific latent heat / J/kg	Energy / J
Lead melting	**2.0**	**23 000**	**46 000**
Ethanol boiling	**0.50**	**850 000**	**425 000**
Aluminium melting	**0.50**	**400 000**	**200 000**

5 CHALLENGE A freezer removes 66 800 J from water at 0 °C. Calculate the mass of ice that forms. Specific latent heat of fusion of water = 334 000 J/kg

$$m = \frac{E}{L} = \frac{66\,800}{334\,000} = 0.20 \text{ kg}$$

INVESTIGATE

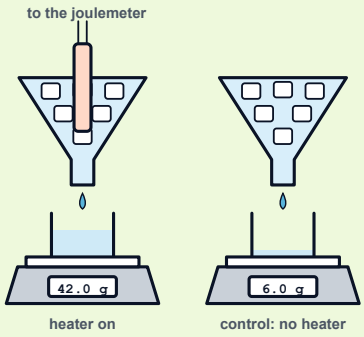
PRACTICAL: MELTING ICE

Measure the specific latent heat of fusion of water.

STAY SAFE

- The heater gets hot: switch it on only when it is surrounded by ice.
- Keep water away from the power supply and wipe up spills.

1. Fill two funnels with the same mass of crushed ice. Put a beaker under each.
2. Put the heater in one funnel. The other is the **control**: it shows how much ice melts just because the room is warm.
3. Switch on the heater and the joulemeter for 5 minutes.
4. Measure the mass of water in each beaker.
5. Mass melted by the heater = mass from the heated funnel − mass from the control.



6. Record your results.

Measurement	Value
Energy supplied (joulemeter) / J	**12 000**
Mass of water from the heated funnel / g	**42.0**
Mass of water from the control funnel / g	**6.0**
Mass melted by the heater / kg	**0.0360**

7. Calculate the specific latent heat of fusion of water from your results. Use $E = m\,L$

EQUATION	$E = m\,L$
REARRANGE	$L = \dfrac{E}{m}$
SUBSTITUTE	$L = \dfrac{12\,000}{0.0360}$
ANSWER	**333 000 J/kg** unit:

8. Explain why the control funnel is needed.

Some ice melts because energy is transferred from the warm room. The control measures this, so it can be subtracted to find the mass melted by the heater alone.

CHECKPOINT

QUICK CHECK

Slide 34, Lessons 5–6. Answer each question, then check it as your teacher reveals the answer.

1

Internal energy is the total of which two energies?

The kinetic energy and potential energy of the particles.

2

What does a flat section of a heating graph show?

A change of state.

3

Which energy of the particles increases while ice melts?

Their potential energy.

4

State the unit of specific latent heat.

J/kg

5

Calculate the energy needed to melt 2.0 kg of ice. L = 334 000 J/kg

$E = 2.0 \times 334\,000 = 668\,000$ **J**

6

Liquid to vapour: latent heat of fusion or of vaporisation?

Vaporisation.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: “Calculate” with a mass in grams: convert to kilograms before you substitute.

9 A 60 g ice cube at 0 °C melts in a drink. Calculate the energy transferred to the ice cube as it melts. Specific latent heat of fusion of water = 334 000 J/kg **[3 marks]**

CONVERT	**m = 0.060 kg (1)**
EQUATION	$E = m\,L$
SUBSTITUTE	$E = 0.060 \times 334\,000$ **(1)**
ANSWER	**20 040 J (accept 20 000 J) (1)** unit:

10 State what is meant by the specific latent heat of vaporisation of a substance. **[2 marks]**

The energy needed to change 1 kg of the substance from liquid to vapour (1) with no change in temperature (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Latent heat is the energy needed for a change of state. The temperature does not change.
- Specific latent heat L: the energy to change the state of 1 kg, in J/kg.
- $E = mL$; rearranged, $L = \frac{E}{m}$ and $m = \frac{E}{L}$.
- Fusion: solid and liquid. Vaporisation: liquid and vapour.
- The same energy is given out when the change is reversed (freezing, condensing).

COVER THE PAGE AND ANSWER

1 Define specific latent heat.

2 Write the equation for the energy for a change of state.

3 State the unit of specific latent heat.

4 Which change of state does “fusion” describe?

I CAN…

- define specific latent heat
- calculate energy, mass or L with $E = mL$
- tell fusion from vaporisation
- describe how to measure the specific latent heat of fusion of water

Lesson

07 SPECIFIC HEAT CAPACITY

AQA Physics 4.3.2.2, 4.1.1.3; Combined Science 6.3.2.2, 6.1.1.3

BIG QUESTION **On a sunny beach the sand burns your feet but the sea is still cold. Both have had the same sunshine. Why?**

BY THE END YOU CAN

- state the three things a temperature rise depends on
- state what specific heat capacity means
- calculate with $\Delta E = m\,c\,\Delta\theta$
- decide whether a problem needs specific heat capacity or specific latent heat

specific heat capacity **temperature change (Δθ)** **thermal energy** **joule per kilogram per degree Celsius (J/kg °C)**

YOU WILL NEED

Per pair: calculator. Demonstration: two beakers on identical hotplates, one with 200 g of water and one with 200 g of cooking oil, each with a thermometer.

Teaching deck, slides 35–39
panphy.app/gcsephy/decks/particle-model/#35

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Write the equation for the energy needed for a change of state.

$E = m\,L$

2
Which energy of the particles increases when the temperature of a substance rises?

Kinetic energy.

3
Water is heated from 18 °C to 63 °C. Calculate the temperature change.

63 − 18 = 45 °C

EXPLAIN

HOW HARD IS IT TO HEAT?

Heat a substance without changing its state and its temperature rises. How much it rises depends on three things:

- the **energy** transferred to it: more energy, bigger rise

- its **mass**: more mass, smaller rise
- the **material** it is made of: the same energy warms 1 kg of copper far more than 1 kg of water

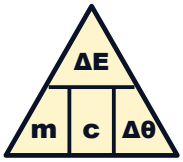
> ! The **specific heat capacity** of a substance is the amount of energy required to raise the temperature of **one kilogram** of the substance by **one degree Celsius**.

$$\Delta E = m\,c\,\Delta\theta$$

change in thermal energy = mass × specific heat capacity × temperature change

ΔE in J, m in kg, c in J/kg °C, $\Delta\theta$ in °C

Rearranged: $c = \dfrac{\Delta E}{m\,\Delta\theta}$ and $\Delta\theta = \dfrac{\Delta E}{m\,c}$. Δ (delta) means “change in”; θ (theta) is temperature.



MATERIAL	SPECIFIC HEAT CAPACITY IN J/KG °C
water	4200
cooking oil	2000
aluminium	900
iron	450
copper	390
lead	130

Water has a very high specific heat capacity: it takes a lot of energy to warm it up, and it gives out a lot as it cools. That is why the sea stays cool on a hot day, and why water is used in radiators and to cool car engines.

WITH YOUR TEACHER SLIDE 37

Calculate the energy needed to heat 2.0 kg of water from 20 °C to 100 °C. Specific heat capacity of water = 4200 J/kg °C

CHANGE	$\Delta\theta = 100 - 20 = 80\ °C$
EQUATION	$\Delta E = m\,c\,\Delta\theta$
SUBSTITUTE	$\Delta E = 2.0 \times 4200 \times 80$
ANSWER	$\Delta E = 672\,000\ J$

YOUR TURN

Calculate the energy needed to heat a 0.50 kg aluminium pan from 20 °C to 50 °C. Specific heat capacity of aluminium = 900 J/kg °C

CHANGE	$\Delta\theta = 50 - 20 = 30\ °C$
EQUATION	$\Delta E = m\,c\,\Delta\theta$
SUBSTITUTE	$\Delta E = 0.50 \times 900 \times 30$
ANSWER	$\Delta E = 13\,500\ J$

FINDING THE TEMPERATURE CHANGE OR THE SPECIFIC HEAT CAPACITY

WITH YOUR TEACHER SLIDE 38

7800 J is transferred to a 0.50 kg block of copper. Calculate its temperature rise. Specific heat capacity of copper = 390 J/kg °C

EQUATION	$\Delta E = m\,c\,\Delta\theta$
REARRANGE	$\Delta\theta = \dfrac{\Delta E}{m\,c}$
SUBSTITUTE	$\Delta\theta = \dfrac{7800}{0.50 \times 390}$
ANSWER	$\Delta\theta = 40\ °C$

YOUR TURN

5400 J is transferred to a 0.20 kg block of aluminium. Calculate its temperature rise. Specific heat capacity of aluminium = 900 J/kg °C

EQUATION	$\Delta E = m\,c\,\Delta\theta$
REARRANGE	$\Delta\theta = \dfrac{\Delta E}{m\,c}$
SUBSTITUTE	$\Delta\theta = \dfrac{5400}{0.20 \times 900}$
ANSWER	$\Delta\theta = 30\ °C$

WITH YOUR TEACHER SLIDE 38

27 000 J raises the temperature of a 1.5 kg metal block by 20 °C. Calculate the specific heat capacity of the metal.

EQUATION	$\Delta E = m\,c\,\Delta\theta$
REARRANGE	$c = \dfrac{\Delta E}{m\,\Delta\theta}$
SUBSTITUTE	$c = \dfrac{27\,000}{1.5 \times 20}$
ANSWER	**c = 900 J/kg °C**

YOUR TURN

12 600 J raises the temperature of 0.60 kg of a liquid by 10 °C. Calculate the specific heat capacity of the liquid.

EQUATION	$\Delta E = m\,c\,\Delta\theta$
REARRANGE	$c = \dfrac{\Delta E}{m\,\Delta\theta}$
SUBSTITUTE	$c = \dfrac{12\,600}{0.60 \times 10}$
ANSWER	**c = 2100 J/kg °C**

Get the marks: $\Delta\theta$ is the **change** in temperature, not the final temperature. From 20 °C to 100 °C is a change of 80 °C. If the question asks for the final temperature, add your $\Delta\theta$ to the starting temperature.

HEAT CAPACITY OR LATENT HEAT?

	SPECIFIC HEAT CAPACITY	SPECIFIC LATENT HEAT
What happens	the temperature changes	the state changes
Temperature	changes	stays constant
Equation	$\Delta E = m\,c\,\Delta\theta$	$E = m\,L$
Unit	J/kg °C	J/kg

PRACTISE

TRY IT

1 Draw a ring around the correct answer in each box.

The unit of specific heat capacity is **J/kg** | **J/kg °C** | **J/°C**

The same energy is given to 1 kg of water and to 1 kg of copper. The bigger temperature rise is in the

water | **copper**

2 SUPPORTED Calculate the energy needed to raise the temperature of 4.0 kg of iron by 25 °C. Specific heat capacity of iron = 450 J/kg °C

EQUATION	$\Delta E = m\,c\,\Delta\theta$
SUBSTITUTE	$\Delta E = 4.0 \times 450 \times 25$
ANSWER	**45 000 J** unit:

3 SUPPORTED 250 g of water is heated from 15 °C to 75 °C. Calculate the energy transferred to the water. Specific heat capacity of water = 4200 J/kg °C

CONVERT	**m = 0.250 kg**
TEMPERATURE CHANGE	$\Delta\theta = 75 - 15 = 60$ °C
EQUATION	$\Delta E = m\,c\,\Delta\theta$
SUBSTITUTE	$\Delta E = 0.250 \times 4200 \times 60$
ANSWER	**63 000 J** unit:

4 Complete the table. Show your working underneath.

Substance	Mass / kg	c / J/kg °C	$\Delta\theta$ / °C	ΔE / J
Cooking oil	**0.50**	**2000**	**60**	**60 000**
Lead	**2.0**	**130**	**20**	**5200**
Unknown metal	**4.0**	**450**	**10**	**18 000**

5 For each change, write the equation you would use.

Change	Equation
Heating water from 20 °C to 80 °C	$\Delta E = m\,c\,\Delta\theta$
Melting ice at 0 °C	$E = m\,L$
Boiling water at 100 °C	$E = m\,L$
Cooling a hot iron bar	$\Delta E = m\,c\,\Delta\theta$

6 CHALLENGE A kettle heats 0.20 kg of water from 20 °C to 100 °C, and then boils all of it away. Calculate the total energy transferred to the water. Use c = 4200 J/kg °C and L = 2 260 000 J/kg

Heating: $\Delta E = m\,c\,\Delta\theta = 0.20 \times 4200 \times 80 = 67\,200$ J. Boiling:

$E = m\,L = 0.20 \times 2\,260\,000 = 452\,000$ J. Total = 519 200 J.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: Work out $\Delta\theta$ first and write it down. It is the step most often missed.

7 A hot-water bottle holds 0.80 kg of water. The water cools from 70 °C to 30 °C. Calculate the energy transferred from the water. Specific heat capacity of water = 4200 J/kg °C **[3 marks]**

TEMPERATURE CHANGE	$\Delta\theta = 70 - 30 = 40$ °C (1)
EQUATION	$\Delta E = m\,c\,\Delta\theta$
SUBSTITUTE	$\Delta E = 0.80 \times 4200 \times 40$ (1)
ANSWER	**134 400 J (1)** unit:

8 A heater transfers 36 000 J to a 2.0 kg block of aluminium. Calculate the temperature rise of the block. Specific heat capacity of aluminium = 900 J/kg °C **[3 marks]**

EQUATION	$\Delta E = m\,c\,\Delta\theta$
REARRANGE	$\Delta\theta = \dfrac{\Delta E}{m\,c}$ (1)
SUBSTITUTE	$\Delta\theta = \dfrac{36\,000}{2.0 \times 900}$ (1)
ANSWER	**20 °C (1)** unit:

9 Give one difference between specific heat capacity and specific latent heat. **[2 marks]**

Specific heat capacity is about a change in temperature (energy to raise 1 kg by 1 °C) (1); specific latent heat is about a change of state at constant temperature (energy to change the state of 1 kg) (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- A temperature rise depends on the energy supplied, the mass and the material.
- Specific heat capacity c: the energy to raise the temperature of 1 kg by 1 °C, in J/kg °C.
- $\Delta E = m\,c\,\Delta\theta$; rearranged, $c = \dfrac{\Delta E}{m\,\Delta\theta}$ and $\Delta\theta = \dfrac{\Delta E}{m\,c}$.
- Temperature changing: specific heat capacity. State changing: specific latent heat.

COVER THE PAGE AND ANSWER

1 Define specific heat capacity.

2 Write the equation for a change in thermal energy.

3 State the unit of specific heat capacity.

4 What does $\Delta\theta$ stand for?

I CAN…

- state what a temperature rise depends on
- define specific heat capacity
- calculate with $\Delta E = m\,c\,\Delta\theta$
- rearrange to find c or $\Delta\theta$
- choose between specific heat capacity and specific latent heat

REFERENCE

EQUATION TOOLKIT

Every equation from Lessons 1–7. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find V from $\rho = \frac{m}{V}$, cover V. m is above ρ, so $V = \frac{m}{\rho}$.

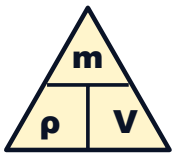
$$\rho = \frac{m}{V}$$

Lesson 3

density = mass ÷ volume

kg/m³, kg, m³ (or g/cm³, g, cm³)

$m = \rho V$ and $V = \frac{m}{\rho}$



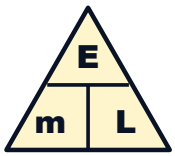
$$E = m\,L$$

Lesson 6

energy for a change of state = mass × specific latent heat

J, kg, J/kg

$L = \frac{E}{m}$ and $m = \frac{E}{L}$



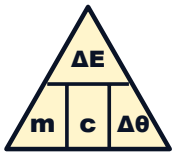
$$\Delta E = m\,c\,\Delta\theta$$

Lesson 7

change in thermal energy = mass × specific heat capacity × temperature change

J, kg, J/kg °C, °C

$c = \frac{\Delta E}{m\,\Delta\theta}$ and $\Delta\theta = \frac{\Delta E}{m\,c}$



1
Write what you know. Convert first: g → kg, cm³ → m³, kJ → J.

2
Write the equation in symbols.

3
Rearrange if you need to, then substitute the numbers.

4
Give the answer with a unit. Ask: is it sensible?

REFERENCE

KEY WORDS FROM THIS BOOKLET

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

Latent heat L6: The energy needed for a substance to change state, with no change in temperature

Specific latent heat (L) L6: The energy needed to change the state of 1 kg of a substance with no change in temperature, in J/kg

Fusion L6: Melting: the change from solid to liquid

Vaporisation L6: The change from liquid to vapour

Specific heat capacity (c) L7: The energy needed to raise the temperature of 1 kg of a substance by 1 °C, in J/kg °C

Temperature change (Δθ) L7: Final temperature minus starting temperature, in °C

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

LESSON 6: SPECIFIC LATENT HEAT

DO NOW

1. The total kinetic and potential energy of all the particles in a system.
2. It stays constant.
3. 0.250 kg

COVER THE PAGE

1. The energy needed to change the state of 1 kg of a substance with no change in temperature.
2. $E = m L$
3. J/kg
4. Solid to liquid (melting).

LESSON 7: SPECIFIC HEAT CAPACITY

DO NOW

1. $E = m L$
2. Kinetic energy.
3. 63 − 18 = 45 °C

COVER THE PAGE

1. The energy needed to raise the temperature of 1 kg of a substance by 1 °C.
2. $\Delta E = m c \Delta \theta$
3. J/kg °C
4. The change in temperature.